Fiscal Year:	FY 2025	**Task Last Updated:**	FY 07/06/2025
PI Name:	Lee, John Ph.D.		
Project Title:	HCAAM VNSCOR: Conversation Analysis to Measure and Manage Trust in Virtual Assistants		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jdlee@engr.wisc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	608-890-3168
Organization Name:	University of Wisconsin, Madison		
PI Address 1:	Department of Industrial and Systems Engineering		
PI Address 2:	1513 University Ave		
PI Web Page:			
City:	Madison	**State:**	WI
Zip Code:	53706-1539	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	04/15/2019	**End Date:**	12/31/2024
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2024 per A. Beitman/HRP (Ed., 6/3/24) NOTE: End date changed to 06/01/2024 per A. Beitman/HRP (Ed., 4/18/23) NOTE: End date changed to 04/14/2023 per S. Huppman/HRP and NSSC information (Ed., 3/20/2020) NOTE: End date changed to 3/31/2020 per NSSC information (Ed., 1/22/2020)		
Key Personnel Changes/Previous PI:	Jerri Stephenson is no longer working in the project. James S. Garrett is now a Co-Investigator.		
COI Name (Institution):	Cross, Ernest Ph.D. (NASA Johnson Space Center) Garrett, James (NASA)		
Grant/Contract No.:	80NSSC19K0654		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

This task is part of the Human Capabilities Assessments for Autonomous Missions (HCAAM) Virtual NASA Specialized Center of Research (VNSCOR).
The goal of this research is to develop conversation analysis to measure and mitigate inappropriate trust in virtual assistants. These trust measurements will guide system design, particularly the multimodal interactions and mode switching, as well as how to mitigate over trust and trust recovery. We will use conversation analysis to measure trust at multiple time-scales from real-time interactions to longitudinal monitoring of trust over a long duration exploration mission.

Conversation analysis provides a promising, but relatively unexplored approach to measuring trust. We propose a conversation analysis at the micro, meso, and macro levels which includes not just the words, but also pauses and facial expressions. Specifically, at the micro-level, conversation elements include voice inflections, pauses between works and keystrokes, gaze shifts, and facial expressions. The meso-level analysis includes words exchanged during interactions with the virtual assistant along with other team interactions as they relate to the automation. At the macro level, conversational analysis considers interaction time, interaction effort, frequency of interaction, turn-taking, barging in tendency, and whether it is the person or the virtual assistant who initiates the interaction. Additionally, prior research into conversational analysis indicates there are novel ways of managing or calibrating trust through the presentation of information, e.g., manipulating the tone and cadence of the system when using speech and through facial expressions (Nass & Brave, 2005; DeSteno et al., 2012).

Due to time delays in communication, long duration exploration missions will require greater crew autonomy and greater reliance on automation. For this approach to work trust calibration needs to be engineered into the system. Trust is a critical construct that mediates how well human operators use automated systems, such as virtual assistants, that provide decision support. Trust affects people's willingness to rely on automated systems in situations that have a degree of uncertainty and risk. Trust strongly affects the effectiveness of human-agent collaboration, particularly in the willingness to accept suggestions from a virtual assistant. Knowing whether or not to trust automation can be further complicated by lack of sleep, workload, task risk, and task complexity. Moreover, as we continue to push the limits of intelligent systems and rely on them more as decision aids trust calibration (i.e., operator trust is at a level which matches the automation's capabilities) becomes essential to mission execution.

Appropriate calibration of trust requires matching the operator's trust to the virtual assistant's current capabilities. Calibration of trust is not something that can happen once, but must occur throughout the life cycle of the interaction between operator and automated system (Hoffman et al., 2009). Trust is a dynamic construct that continuously increases and decreases due to a number of factors, primary the performance of the automated system, i.e., higher performance leads to higher trust and vice versa. Although much effort focuses on creating more capable and trustworthy automation, less effort has considered the equally important consideration of creating trustable automation. Trustable automation is automation that is understandable and that naturally promotes calibrated trust. Therefore, we aim to create trustable automation by continuously measuring operators' trust unobtrusively and in real-time, and then use this measure to guide the virtual agent to employ one or more countermeasures to calibrate trust and improve human-system performance.

References

DeSteno D, Breazeal C, Frank RH, Pizarro D, Baumann J, Dickens L, Lee JJ. Detecting the trustworthiness of novel partners in economic exchange. Psychol Sci. 2012 Dec;23(12):1549-56. http://doi.org/ ; PubMed PMID: 23129062

Hoffman RR, Lee JD, Woods DD, Shadbolt N, Miller J, Bradshaw JM. The dynamics of trust in cyberdomains. IEEE Intelligent Systems. 2009 Nov-Dec;24(6):5-11. https://

Nass C, Brave S. Wired for Speech : How Voice Activates and Advances the Human-Computer Relationship. Cambridge, MA: MIT Press, 2005.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The outcomes of this research will make two important contributions to the overall HCAAM VNSCOR effort. First, it will promote more effective interactions and acceptance of virtual assistants. Second, it will provide new analytic techniques for understanding how people work with automated agents as team members.
Virtual assistants and other types of agents enabled by artificial intelligence represent an important opportunity to extend human capabilities, but only if they are accepted and trusted appropriately. If people trust the virtual assistant too much they will rely on it in situations that exceed its capability, and if they trust it too little they will fail to engage it when it could benefit the team. One pathway towards appropriate trust is to make the virtual assistant more trustworthy: increase its technical capabilities to accommodate any situation. Another approach is to make it more trustable: communicate its capability and allow its capability to be challenged in its interactions with people. Such trustable technology requires three important advances to the state of knowledge in the field:

1. An ability to ascertain how much people currently trust the technology

2. An ability to convey uncertainty and its capability, particularly as part of conversational interactions

3. Interaction affordances that provide the opening for people to assess the capability of the assistant, particularly as part of conversational interactions.

These three advances for trustable technology require the development of new analytic techniques for understanding human interaction with automated teammates. Real-time, unobtrusive measures of trust represent a particularly valuable, but challenging measure to develop. Trust is most often measured with ratings and indirectly through people's decision to rely on automation, which are obtrusive not diagnostic. Conversation and text-based interactions offer a promising, but unexplored way to assess trust. Text analysis has a 50-year history in domains as diverse as psycholinguistics and cognitive science, and more recently natural language processing, affective state assessment, and sentiment analysis. Building on the foundation of text analysis makes it possible for this research to immediately contribute to data analysis of previous and future studies of automation-human teaming, and to contribute to the foundation of conversational agent design.

Task Progress:	This research explored the complex, non-linear dynamics of trust in human-automation teams, challenging the traditional assumption that trust stabilizes to predictable levels over time. Drawing from evolutionary biology's concepts of convergent and contingent events, the work demonstrated that small, seemingly insignificant changes in team interactions could lead to dramatically different trust trajectories—much like Stephen J. Gould's theory that replaying evolution would rarely produce the same outcomes. In human-automation systems, these contingent dynamics were found to cause groups to gravitate toward extreme, bi-modal trust levels, making traditional statistical analysis inadequate for understanding trust evolution in complex, long-duration missions. The core empirical work centered on longitudinal studies of teams that worked with automated systems over multiple days, addressing a significant gap in human-automation research where most studies had spanned only hours rather than extended periods. Through analysis of naturalistic team conversations and trust measurements, researchers quantified how day-to-day group interactions influenced individual trust development in automated systems. They calculated network distances between team members based on interaction patterns and compared these with trust differences within teams. While the quantitative analysis revealed statistically significant but weak correlations between network distance and trust differences, the qualitative assessment showed clear evidence of trust divergence that persisted over time, suggesting that team conversation structures and seemingly irrelevant interactions played crucial roles in shaping trust trajectories. To formalize these findings, the research developed computational models that treated trust as a stochastic differential process influenced by both direct automation interactions and indirect team conversations. This approach introduced the concept of "meta-trust"—trust in teammates' trust judgments about automation—extending individual-focused models to capture team-level dynamics. The models demonstrated how team conversation structures, turn-taking patterns, and social feedback mechanisms affected trust diffusion, potentially leading to divergent team trust states rather than convergence to common values. Comprehensive sensitivity analyses examined how initial trust levels, inertia, and information weighting shaped trust evolution, revealing that high inertia or low weighting led to minimal trust change while team conversation weighting modulated trust alignment or divergence. Building on statistical physics principles, the research employed the Potts model to conceptualize trust as a socially modulated phenomenon within teams, formalizing how team members' trust levels influenced each other through opinion alignment and consensus formation processes. This approach emphasized distributed cognition, where teams collectively calibrated trust in response to automation reliability, demonstrating that trust was co-constructed rather than individually maintained. The analysis of NASA HERA naturalistic data revealed that during periods of automation unreliability, trust alignment increased as teams shifted toward social consensus for uncertainty resolution, suggesting that trust was adaptively regulated based on both automation performance and social interaction dynamics. The methodological innovations included adapting the Potts model to quantify trust alignment and coupling strength between teammates, incorporating phase-specific multipliers to account for changing trust dynamics under different automation reliability conditions. Using Bayesian hierarchical modeling with weekly trust measures, the research provided quantitative evidence that social interaction increased trust alignment, particularly under low reliability conditions, while capturing nonlinear, emergent, and collective trust dynamics that traditional methods could not adequately address. These findings revealed that trust contagion could both stabilize and destabilize teams, with trust cascades potentially arising from uncertainty rather than automation failures per se. To support analysis of rich conversational data, the research developed an innovative interactive data visualization tool using RShiny and UMAP (Uniform Manifold Approximation and Projection) to help practitioners quickly identify critical events in dialogue data. This tool addressed the challenge of moving from standard numerical data to complex conversational databases by providing a two-dimensional visualization of utterances that preserved both local and global structures of high-dimensional conversational space. The interface allowed analysts to explore affective dialogue information across time through slider features, offering a top-down approach that contrasted with typical bottom-up qualitative analysis methods and reduced the cognitive demand of switching between multiple interfaces. The implications for AI teaming design were profound, suggesting that trust calibration strategies must account for potential divergence within and across teams rather than simply aiming for convergence to appropriate trust levels. System designs should incorporate mechanisms to monitor and mediate trust diffusion through team conversations, particularly in long-duration, high-stakes missions. The research recommended structuring communication to avoid premature consensus, balancing anthropomorphism with epistemological transparency, and recognizing that trust dynamics were fundamentally social phenomena that emerged from team interactions rather than individual assessments alone. This work fundamentally extended conceptual models of trust from individual to team dynamics by incorporating meta-trust and stochastic diffusion processes, providing both theoretical frameworks and practical methodologies for understanding and optimizing trust dynamics in human-AI collaborative systems. The research demonstrated that trust in automation was not merely an individual psychological phenomenon but a complex social process shaped by team interactions, contingent events, and emergent dynamics that required sophisticated modeling approaches to understand and predict. These insights proved crucial for designing resilient human-AI teams that could maintain appropriate trust calibration despite the inherent uncertainties and complexities of real-world collaborative environments.
Bibliography Type:	Description: (Last Updated: 01/21/2026)
Abstracts for Journals and Proceedings	Lee JD, Renata V, Klaeser M, Cross E, Garrett J. "Dynamics of trust in automation in team-based networks." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Articles in Peer-reviewed Journals	Alsaid A, Li M, Chiou, EK, Lee JD. "Measuring trust: A text analysis approach to compare, contrast, and select trust questionnaires." Front Psychol. 2023 Nov 15;14:1192020. https://doi.org/10.3389/fpsyg.2023.1192020 , Nov-2023
Articles in Peer-reviewed Journals	Klaeser M, Renata V, Lee JD. "Unlocking insights: An interface for understanding dialogue data." Proc Hum Factors Ergon Soc Annu Meet. 2024;68(1):336-7. https://doi.org/10.1177/10711813241263505 , Aug-2024
Articles in Peer-reviewed Journals	Li M, Erickson IM, Cross EV, Lee JD. "Estimating trust in conversational agent with lexical and acoustic features." Proc Hum Factors Ergon Soc Annu Meet. 2022 Oct;66(1), 544-8. https://doi.org/10.1177/1071181322661147 , Oct-2022

Articles in Peer-reviewed Journals	Li M, Erickson IM, Cross EV, Lee JD. "It's not only what you say, but also how you say it: A machine learning approach to estimate trust from conversation. " Hum Factors. 2024 Jun;66(6):1724-41. https://doi.org/10.1177/00187208231166624 ; PubMed PMID: 37116009; PubMed Central PMCID: PMC11044523 , Jun-2024
Articles in Peer-reviewed Journals	Li M, Lee JD. "Purpose outweighs performance: Trust drops more when AI teammates fail to cooperate, but explanations can repair it." International Journal of Human-Computer Interaction. 2025 Jul 31:1-22. https://doi.org/10.1080/10447318.2025.2539449 , Jul-2025
Articles in Peer-reviewed Journals	Renata V, Klaeser M, Lee JD. "The relevance of seemingly irrelevant group interactions for automation trust: An analysis of naturalistic team conversations on contingent behaviors." Proc Hum Factors Ergon Soc Annu Meet. 2024 Aug 29;68(1), 302-4. https://doi.org/10.1177/10711813241262023 , Aug-2024